From Frame to Episode: Evidence-Preserving Alert Mediation for Selective Inference in Vehicular Edge Systems

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***Abstract*—Vehicular edge nodes often reduce per-frame computation to meet latency and energy constraints. This strategy, however, can remove information that the alerting layer still needs to determine whether an alert should be delivered. When frames are skipped and previous predictions are reused, the system may no longer know whether the available evidence is sufficiently current to support an alert. We present a cognitive edge architecture that separates *when to compute* from *when to alert* across four decision levels and makes suppression itself first-class, auditable evidence. A Suppression Summary produced by the Operational State and Suppression Layer (OSSL) records information about every skipped frame at constant cost, without inferring a new risk estimate. An Episodic Alert Mediation Layer (EAML) is the single origin of driver alerts and requests fresh computation whenever high episodic risk is supported by reuse-dominated evidence. The architecture was evaluated in a controlled synthetic vehicular-risk environment using separate development seeds and an internally frozen 20-seed holdout. First, we compared the proposed architecture with a legacy frame-level trigger and an event-governed safeguard baseline that does not use suppression evidence. The full mediation satisfied the complete pre-declared admissibility specification. Event recall increased by 0.021 (95% CI $[0.006, 0.037]$) with effective-latency parity, false alerts per trip decreased by 62%, and the stale-alert rate decreased from 0.030 to zero. Removing the Suppression Summary made the design inadmissible and reintroduced stale alerts (+0.0066, CI $[0.005, 0.008]$). Together with the mechanism-decomposition analysis, this supports mechanism-level attribution under the evaluated simulation assumptions. Second, under an oscillating-risk stress regime, the legacy trigger produced 13.1 alerts per trip, while the event-governed baseline lost approximately two thirds of its recall. The mediated architecture retained substantially more coverage than the governed baseline and maintained zero stale alerts, although no evaluated arm satisfied the alert budget in this stress regime. Finally, post-hoc analyses separated the effects of evidence preservation and forced refresh, and showed that the results are not artifacts of the stale-alert accounting threshold or of baseline calibration. These results validate the architectural mechanism under controlled simulation assumptions; evaluation with naturalistic driving data and embedded hardware remains future work.**

***Index Terms*—edge–cloud continuum, edge AI, selective computation, alert governance, self-adaptive systems, evidence architecture, vehicular safety, reproducibility**

I. Introduction

Real-time driving-risk detection at the edge must balance three requirements: low latency, limited resource use, and reliable communication with the driver [1], [2]. Selective computation helps satisfy the first two by suppressing redundant model updates and reusing previous predictions [3], [4]. Alert delivery, however, is decided by a separate mechanism. This creates an important interaction between computation and communication: the alerting layer may depend on information that selective computation removes. When a gate skips a frame, the skipped instant often leaves no record beyond a reused probability. Downstream components may therefore be unable to distinguish a newly computed risk estimate from one that has been reused across several frames without re-evaluation.

This observation motivates the design question addressed in this work: what information must be preserved when computation is skipped so that alert decisions remain well grounded? We focus on two failure modes. The first arises when frame-level alert triggers are combined directly with temporal governance mechanisms such as thresholds, cooldowns, and re-arming. In this setting, the trigger controls when an alert is requested, whereas the safeguard controls whether and when it is actually delivered. As a result, fast threshold crossing does not necessarily imply timely or appropriate driver notification. The second failure mode arises when skipped computation leaves no explicit evidence about the quality of the reused state. The alerting layer can then act on evidence whose freshness it cannot assess. We refer to alerts issued from evidence windows dominated by reused computation as *stale alerts*. Improving the detector alone does not address either problem because both concern where and how alert decisions are made.

We address these failure modes with a cognitive edge architecture organized into four decision levels (Fig. 1). The levels separate four questions that are often handled by a single threshold-based pipeline. Level 1, the Multimodal Hybrid Execution Gate (MHEG), decides whether the detector should execute on the current frame. Its decision is based on input variation and predictive uncertainty, and it is limited to computation scheduling. Level 2, the Operational State and Suppression Layer (OSSL), records what remains known when computation is skipped. Every frame produces exactly one evidence object: either a computed state z_t or a Suppression Summary OSS_t. The skipped computation is therefore explicitly represented rather than silently discarded. Level 3, the Episodic Alert Mediation Layer (EAML), decides whether accumulated evidence is sufficient to deliver an alert. It is the single origin of driver alerts, aggregates z_t and OSS_t over an evidence window, computes episodic risk, and applies

an event-governed delivery rule. When high episodic risk is supported by evidence dominated by reuse, the EAML requests fresh computation before deciding whether to alert. Level 4, the cloud, governs slower configuration changes for future operation. It may promote new policy parameters but does not bypass local mediation or trigger driver alerts directly.

The paper makes three contributions. First, we introduce an *evidence-preserving architecture* for edge alert governance with explicit invariants at each decision level. Its central requirement is that skipped computation must leave auditable evidence, at constant per-frame cost, for the component responsible for communication. Second, we define the *stale-alert rate* (SAR), an operational metric that measures the fraction of delivered alerts whose evidence window is dominated by reused computation. Third, we provide a confirmatory evaluation of the architecture using a protocol frozen before a 20-seed holdout. The evaluation includes a documented amendment trail in which two earlier specifications produced empty admissible sets, an ablation designed to test the role of suppression evidence, and an oscillating-risk stress analysis that examines behavior outside the nominal evaluation regime.

The evaluation is organized around three research questions that move from overall effect to mechanism and then to behavior under stress. *RQ1*: Does episodic mediation improve alert quality over an event-governed safeguard baseline without sacrificing recall or effective latency, while remaining within a declared alert budget? *RQ2*: Does suppression evidence contribute to that improvement, or can the mediation succeed without it? *RQ3*: How do the competing designs behave under an oscillating-risk regime that stresses the interaction between selective computation and alert delivery?

II. Related Work

The design space addressed in this work lies at the intersection of four research areas: selective computation, streaming risk detection, runtime assurance, and information freshness. Each contributes part of the solution, but none directly addresses how alerting should reason about evidence that was not freshly computed.

Selective and adaptive computation. Skip RNN [3], adaptive computation time [4], adaptive frame selection [5], and coarse-to-fine inference [6] reduce the cost of recurrent or video models by suppressing unnecessary updates or selecting more informative inputs. Their primary objective is computational efficiency while preserving task performance. In these approaches, however, a skipped instant is typically represented only through the reuse of an earlier state or prediction. The skipped computation itself does not produce an explicit evidence object that records what was omitted, how long the reused state has persisted, or how the input evolved during that period. MOSAIC [20] is a closer point of comparison because it tracks prediction age and reuse across concurrent video streams for criticality-aware scheduling. That information is used to support resource allocation, rather than downstream alert decisions. In our architecture, selective computation remains a scheduling mechanism at the first level, while the downstream consequences of suppression are represented explicitly for later decision making.

Streaming risk detection and online action detection. Causal recurrent methods for driver monitoring [8], [9] and online action detection [7] focus on producing reliable predictions under strict temporal causality, usually evaluated with detection-oriented aggregate metrics. A separate body of work on collision-warning systems shows that the usefulness of a warning also depends on when it is delivered and on how often false or redundant alerts occur [14], [15]. These results motivate evaluating the *delivered alert*, rather than the frame-level detection score alone. What is generally not modeled in this literature is the interaction between alert delivery and selective computation, especially when an alert decision depends on predictions that may have been reused across skipped frames.

Runtime assurance and self-adaptive systems. Runtime assurance architectures mediate proposed actions before execution [12], [13], while autonomic and self-adaptive systems organize adaptation through feedback loops such as MAPE-K over shared knowledge [10], [11]. These perspectives motivate two boundaries in our architecture. Decisions that affect the driver are mediated locally at the edge, whereas slower configuration changes are handled separately through governed policy promotion in the cloud. The remaining question is what information the local alerting layer can rely on when selective computation means that many frames were never freshly evaluated. Our work addresses this within-edge evidence problem rather than proposing a new runtime assurance or self-adaptation mechanism.

Information freshness and age-aware systems. Age-of-information (AoI) research studies the timeliness of updates and its effect on decisions in networked and edge systems [18]. Related work has examined how stale observed features affect real-time inference accuracy [19] and how peak-age objectives can guide scheduling or forced updates [21]. These approaches reason primarily about the age of data or updates. Our concern is different: an alert may be delivered at the current time while still being supported mostly by reused inference evidence. The stale-alert rate (SAR) therefore measures the composition of the evidence window that supports a delivered alert, rather than the transport age of the alert or the age of the latest update. Freshness-aware sensing provides a related mechanism by suppressing or requesting sensor updates according to age or uncertainty [22]. In that setting, freshness control acts on sensing and transmission. Here, freshness is evaluated at the alert-decision layer, which may request fresh inference before a driver notification is delivered.

Taken together, these lines provide the individual mechanisms needed for selective execution, temporal detection, safeguarded delivery, and freshness-aware control. In the literature reviewed here, however, including a structured cross-domain audit of the four lines above, we did not find an evaluated architecture that combines a per-frame evidence contract for suppressed computation with episodic alert mediation under an explicit alert budget. The novelty is not a new gating

TABLE I
POSITIONING OF REPRESENTATIVE MECHANISMS.

Mechanism	SE	EA	DS	AB	SA
Skip RNN / ACT [3], [4]	–	–	–	–	–
AdaFrame / LiteEval [5], [6]	–	–	–	–	–
MOSAIC [20]	△	–	–	–	–
Runtime assurance [12], [13]	–	–	✓	–	–
AoI / freshness [18], [19], [21]	–	–	–	–	△
Freshness-aware sensing [22]	–	–	–	△	△
Event-governed safeguard	–	event	✓	✓	–
This work (OSSL+EAML)	✓	✓	✓	✓	✓

SE: suppression evidence; EA: episodic alerting; DS: delivery safeguard; AB: alert budget; SA: staleness audit. △ denotes a related notion without an explicit suppression-evidence contract for delivered-alert mediation. The event-governed safeguard is the frozen A1 baseline, not prior work.

rule, age metric, safeguard, or forced-update mechanism in isolation. It is a per-frame evidence contract that makes skipped computation explicitly observable to downstream decision layers, combined with two architectural invariants: every frame produces exactly one evidence object, and every driver alert originates from evidence-aware episodic mediation. Fresh inference may be requested when the evidence supporting a candidate alert is dominated by reused computation. Table I summarizes this positioning.

III. ARCHITECTURE: FOUR DECISION LEVELS

The gaps identified above motivate separating four decisions that are often handled by a single threshold-driven pipeline: when to compute, what evidence to retain when computation is skipped, when to deliver an alert, and when to change the operating configuration. We consider a strictly causal stream of frames $x_t \in \mathbb{R}^D$ containing kinematic, behavioral, and contextual attributes. When executed, a recurrent detector [23] produces a risk probability p_t. Consecutive frames are grouped into episodes of length T, and critical episodes contain a risk onset instant t_{onset}.

Figure 1 shows how these decisions are separated across the four levels.

A. Level 1 (MHEG): when to compute

The MHEG is responsible only for computation scheduling. Its question is whether the current frame contains enough new information to justify running the detector. The detector executes when the input variation satisfies $|\Delta x_t| \geq \tau_\Delta$ over a kinematic subset or when the previous prediction is sufficiently uncertain, $H(p_{t-1}) \geq \tau_H$. A hard limit, max_skip, bounds the number of consecutive suppressed frames. The MHEG also exposes a *force channel* through which higher levels can request an additional detector execution when fresh evidence is needed.

These decisions are binary: execute the detector or reuse the most recent computed state. The MHEG remains external to the detector and does not participate in alert delivery. In the evaluated configuration, τ_Δ=0.005, τ_H=0.6, and max_skip=8, which suppresses approximately 68% of frames.

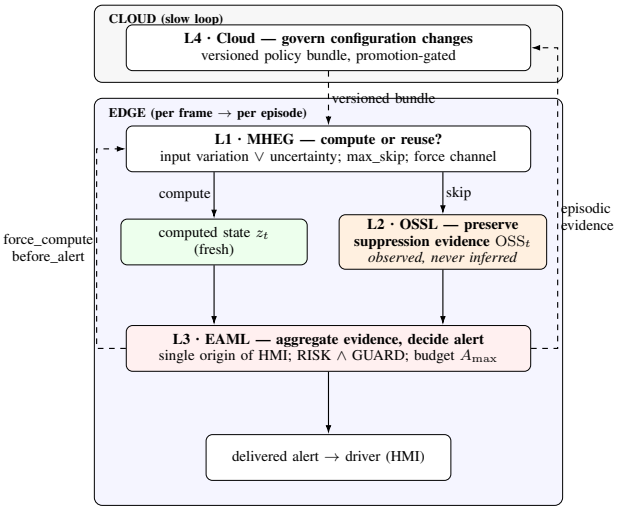


Fig. 1. Four decision levels. Every frame yields exactly one evidence object (z_t or OSS_t). The EAML is the only path to the driver's human–machine interface (HMI), and the cloud changes parameters only through governed promotion.

TABLE II
THE SUPPRESSION SUMMARY OSS_t (PER SUPPRESSED FRAME, $O(1)$).

Field	Meaning
reused_from, elapsed	index and age of the last computed state
Δx, $\max \Delta$	accumulated and maximum input variation since that state
entropy_carried	uncertainty of the reused prediction
confidence_decay	age- and variation-based decay estimate
cheap_risk_proxy	low-cost kinematic risk indicator
trigger_margins	distances to τ_Δ and τ_H at skip time
suppression_count	length of the current suppression run
quality	derived tag in {safe, watch, unsafe}

B. Level 2 (OSSL): suppression as first-class evidence

Selective computation creates an information gap whenever the detector is not executed. The Operational State and Suppression Layer (OSSL) makes that gap explicit through a per-frame evidence contract. Every frame produces exactly one evidence object: either a freshly computed state z_t, containing p_t and its uncertainty, or a Suppression Summary OSS_t.

The Suppression Summary does not estimate a new risk probability. Its purpose is to describe the skipped computation well enough for downstream components to assess the quality of the reused state. It therefore stores, in constant space, the source of the reused state and its age, accumulated and maximum input variation, carried entropy, a confidence-decay estimate, a low-cost risk proxy, the margins to the two gate thresholds, the current suppression-run length, and a derived quality tag in $\{$safe, watch, unsafe$\}$.

Three invariants define the OSSL. (I1) OSS_t never creates a new risk probability; suppressed computation is observed and summarized rather than inferred. (I2) Each summary requires $O(1)$ work and storage per frame. (I3) Every frame contributes an evidence object, so the evidence stream contains no gaps. Together, these invariants make suppression explicit and auditable for downstream mediation and analysis. Table II lists the complete Suppression Summary.

C. Level 3 (EAML): when to alert

The Episodic Alert Mediation Layer (EAML) is the only component allowed to deliver driver alerts. A high frame-level detector score is therefore treated as evidence for an alert decision rather than as an alert by itself. The EAML processes the evidence in a fixed sequence. It first aggregates the current evidence window, then estimates episodic risk, applies persistence and escalation logic, evaluates the event-governed guard, checks evidence freshness, and only then decides whether an alert should be delivered.

For a sliding window W of evidence objects, episodic risk is

$$R_E = 0.7 \cdot \overline{\text{top-}k} + 0.3 \cdot \text{peak},$$

where $\overline{\text{top-}k}$ is the mean of the k highest risk values in the window and *peak* is the maximum. Delivery requires both a RISK condition and a GUARD condition. RISK is satisfied by persistence of m frames above θ_{ep}, by a peak condition, or by escalation. GUARD is implemented by the event-governed safeguard (EGS). An event opens at θ_{on}, closes below θ_{off}, and can generate another alert only after escalation or lack of driver response, subject to a minimum gap. The per-trip limit $A_{\max}$ bounds the total number of delivered alerts.

Three invariants govern this level. (E1) No driver alert is generated outside the EAML. (E2) A single high-risk frame cannot satisfy the RISK condition on its own. (E3) If episodic risk is high while more than 60% of the evidence window is reused and the latest detector execution is sufficiently old, the EAML issues *force_compute_before_alert*. The MHEG then performs the requested computation, after which the EAML re-evaluates the alert decision using the refreshed evidence. This prevents the system from delivering an alert solely on the basis of a reuse-dominated evidence window.

D. Level 4 (cloud): adaptation by promotion only

Level 4 handles slower system adaptation. The parameters τ_Δ, τ_H, W, m, θ_{ep}, the guard settings, and $A_{\max}$ are stored in a versioned policy bundle.

The cloud may propose a new bundle, but that bundle reaches the edge only through a governed promotion decision based on pre-declared criteria over delivered outcomes. The cloud does not generate driver alerts and does not override the EAML. Level 4 defines an architectural boundary rather than an evaluated component. Cloud-side policy promotion is neither implemented nor experimentally evaluated in this study; it is included to specify that remote configuration changes cannot bypass local alert mediation.

E. Threshold provenance and freezing

The thresholds used by the architecture serve different purposes and were not all established in the same way. Parameters that control the mediated alerting policy were selected using development data only and were frozen before the confirmatory holdout. In particular, the EAML configuration (W, m, θ_{ep}, the guard thresholds, and $A_{\max}$) was selected from the development search described in Sec. IV and then held fixed for confirmatory evaluation.

The MHEG parameters τ_Δ, τ_H, and max_skip define the selective-computation regime used throughout the study. They were fixed during development to establish the evaluated suppression operating point and were not tuned using holdout outcomes.

The reuse threshold of 0.60 serves a different role. It defines a reuse-dominated evidence window as one in which a strict majority of the evidence has been reused. The threshold was fixed during development and was not selected using holdout results. Its sensitivity is examined post-hoc at 0.50, 0.60, and 0.70.

These architectural thresholds are distinct from the admissibility criteria of Sec. IV, which are declared engineering criteria rather than universal human-factors constants. No architectural or evaluation threshold was re-optimized using the confirmatory holdout.

F. Operational metrics

The evaluation measures five operational properties of the alerting system: coverage, timeliness, false-alert burden, evidence freshness, and total alert load.

Event recall is the fraction of critical episodes for which at least one alert is delivered after the risk onset. *Effective time-to-alert* combines detection delay and missed-event penalty:

$$\text{TTA}_{\text{ef}} = (t_{\text{alert}} - t_{\text{onset}}) \cdot \Delta t \; + \; \mathbf{1}[\text{miss}] \cdot T_{\max}, \quad (1)$$

where $T_{\max}$=10 s. A critical episode that receives no alert is therefore assigned the maximum delay. The *false-alarm rate* (FAR) counts alerts delivered during non-critical episodes per trip. The total number of *alerts/trip* is evaluated against the declared budget B=2.5.

Evidence freshness is measured by the *stale-alert rate* (SAR). For the set $\mathcal{A}$ of delivered alerts,

$$\text{SAR} = \frac{1}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} \mathbf{1}[\rho_W(a) > 0.60]\,, \quad (2)$$

where $\rho_W(a)$ is the fraction of reused evidence in the decision window when alert a is delivered; the provenance of the 0.60 threshold is described above.

Lower SAR indicates that fewer delivered alerts depend on reuse-dominated evidence. For example, SAR=0.03 means that 3% of delivered alerts were issued from windows classified as stale, whereas SAR=0 means that no delivered alert met this criterion. SAR should not be interpreted as the age of the alert itself. It measures how much of the evidence supporting the alert was reused rather than freshly computed.

Redundant alerts per trip, defined as repeated alerts within an event that has already been notified, are reported as an additional quality measure. Accordingly, higher event recall is preferred, while lower TTA_{ef}, FAR, SAR, and redundant-alert counts are better. Alerts per trip is treated differently: it is an operational cost that must remain within the declared budget.

IV. Frozen Holdout Protocol

The metrics defined above determine the acceptance logic of the experiment. A candidate is considered useful only if it preserves coverage and responsiveness, remains within the declared alert budget, does not worsen false-alert burden, and improves at least one alert-quality dimension. These requirements were encoded in an internally frozen holdout protocol finalized before any inspection of the holdout. The protocol was not externally preregistered.

All experimental arms use the same episode generator, observation model, MHEG configuration, and pre-drawn exogenous realizations, verified by assertion; the arms differ only in the alert-decision layer.

A. Experimental arms

Each arm addresses a different experimental question. A0 represents a conventional unbudgeted frame-level trigger. A1 adds event-governed delivery but does not use suppression evidence. A2 instantiates the complete architecture described in Section III. A3 removes the suppression-evidence mechanism from A2 and is used as the primary ablation. A4 is a post-hoc decomposition arm that retains suppression evidence but disables forced refresh.

A0 - Legacy. A frame-level trigger fires when $p_t \geq \theta$, with cooldown 3 and hysteresis 0.05. It has no alert budget and serves as the descriptive as-is reference.

A1 - EGS. The event-governed safeguard baseline uses θ_{on}=0.55, θ_{off}=0.52, gap 2, and $A_{\max}$=3. Its configuration predates this study and was kept frozen rather than re-optimized for the confirmatory experiment. A1 governs alert delivery at the event level, but it is not guaranteed to satisfy the alert budget adopted in this study. Because A1 predates the current experiment and was not optimized for its alert budget, conclusions from the primary A2–A1 comparison are interpreted jointly with the re-optimized safeguard analysis reported in Section V-D.

A2 - EAML. The complete mediated architecture described in Section III.

A3 - EAML−OSSL. This ablation is identical to A2 except that suppressed frames enter the evidence window without OSS_t. The previous probability is carried forward without the associated quality or decay information, and the force channel is disabled.

A4 – EAML with OSSL and no forced refresh. This post-hoc mechanism-decomposition arm retains the Suppression Summary but disables the forced-refresh channel. It is not included in the pre-declared admissibility analysis and is used only to separate the effects of evidence preservation and freshness restoration.

B. Scenarios and experimental design

Each episode contains T=96 frames (≈10 s) and D=31 attributes. Episodes are generated from behavioral recipes covering normal (N01–N03), attention (A01–A03), borderline (L01–L03), abrupt-critical (C01–C04), and progressive-critical (C05–C10) patterns. The borderline recipes are the primary source of false alerts. Feature and risk calibration are derived from naturalistic-driving statistics [16].

Twelve synthetic driver personas represent distinct behavioral profiles, each defined by three modifiers: risk exposure, responsiveness to alerts, and fatigue, which degrades that response. The personas exist so that the results do not depend on a single behavioral profile. Each persona drives 20 trips, and each trip is a sequence of 12 episodes, so one seed contains $12 \times 20 \times 12 = 2{,}880$ episodes; a seed is one complete repetition of the simulated world with different pre-drawn random realizations. The fixed recipe mix of 50/15/15/10/10% determines how these 2,880 episodes are distributed over the five families: 1,440 normal, 432 attention, 432 borderline, 288 abrupt-critical, and 288 progressive-critical episodes.

This design fixes the evaluation denominators. Each seed contains 576 critical episodes (the abrupt and progressive families, 20% of the mix), over which event recall is computed, and 2,304 non-critical episodes, over which false alerts are counted; the 20 holdout seeds therefore comprise 57,600 episodes. The trip is the operational unit: alert load and FAR are reported per trip, and the total load is compared with the declared budget B. Because the recipe mix is fixed and all exogenous realizations are pre-drawn and shared, every arm faces exactly the same episode sequence, so outcome differences are attributable to the alert-decision layer alone.

The observation model is declared rather than learned. Observed risk is

$$p_{\text{obs}}(t) = \text{clip}(y(t) + \varepsilon_t \cdot 0.10 \cdot (1.2 - \omega)),$$

where $y(t)$ is the latent risk defined by the recipe, $\varepsilon_t \sim \mathcal{N}(0,1)$ is pre-drawn, and ω is the persona-specific observability parameter. The entropy supplied to the MHEG is computed from the most recent detector prediction. Driver response follows

$$q = \text{clip}(\text{resp} - 0.4 \cdot \text{fatigue}, 0.05, 1),$$

with the same pre-sampled uniform realizations used for every arm.

Statistical attribution is performed at the persona×seed cluster level. Paired differences are aggregated per cluster before inference [17]. All intervals were computed from the 240 paired persona×seed cluster differences using a normal approximation, the mean difference ± 1.96 standard errors for the two-sided 95% intervals, with the standard error estimated from the sample variance of the cluster differences. The persona×seed cluster, rather than the individual episode or alert, was the unit of inference. Stochastic episode realizations (recipe selection, observation noise, and driver-response draws) were distinct for every persona×seed combination and never reused across combinations; common random numbers were shared only across experimental arms within the same combination. U90 denotes the upper endpoint of the corresponding two-sided 90% interval (mean + 1.645 SE), used by the admissibility gates as a conservative one-sided upper bound.

Development used seeds 201–205 to select the EAML operating configuration from a grid of 162 candidates over window length, persistence, episodic threshold, and guard parameters. The frozen configuration is $W{=}8$, $m{=}3$, $\theta_{\text{ep}}{=}0.70$ with guard parameters $\theta_{\text{on}}{=}0.65$, $\theta_{\text{off}}{=}0.62$, gap 2, and $A_{\max}{=}4$. The 20 holdout seeds, corresponding to 240 persona×seed clusters, were generated once, remained uninspected during development, and were executed exactly once after the configuration was frozen.

C. Admissibility gates and amendment trail

The admissibility specification translates these operational objectives into explicit decision criteria. Relative to the governed baseline A1, a candidate is admissible only if all of the following conditions hold simultaneously:

$$\begin{aligned}
&\text{G1}\ \ \text{U90}(\Delta\text{TTA}_{\text{ef}}) < +1.0\,\text{s} \quad \text{(non-inferior latency)}\\
&\text{G2}\ \ \Delta\text{Recall} \geq -0.02 \quad \text{(non-inferior coverage, per regime)}\\
&\text{G3}\ \ \text{Alerts/trip} \leq 2.5 \quad \text{(absolute budget)}\\
&\text{G4}\ \ \text{U90}(\Delta\text{FAR}) \leq 0 \quad \text{(no false-alert regression)}\\
&\text{G5}\ \ \text{superiority in} \geq 1 \text{ of } \{\text{SAR}, \text{redundant}, \text{FAR}\}.
\end{aligned}$$

G2 is evaluated both in aggregate and separately for abrupt and progressive onset regimes. G5 requires the corresponding 95% confidence interval to exclude zero. The three G5 dimensions are treated as a joint operational admissibility assessment rather than as three independent confirmatory hypotheses; their individual confidence intervals are reported transparently and interpreted together. An empty admissible set is treated as a valid experimental outcome.

The admissibility specification was revised twice using development data only, before holdout inspection. Version 1 compared the budget-constrained candidates with the unbudgeted A0 reference and yielded no admissible candidate among 54 configurations. Version 2 required strict latency superiority over A1 and yielded no admissible candidate among 162 configurations. Version 3 replaced strict latency and recall superiority with non-inferiority requirements while retaining the budget and alert-quality gates. The 1.0-s latency margin was selected after development-data inspection. All Version 3 criteria and the selected operating configuration were frozen before the 20-seed holdout was inspected. The complete amendment trail is preserved in the artifact.

V. Results

A. RQ1: confirmatory holdout evaluation

RQ1 asks whether the full mediation improves alert quality over the governed baseline without losing coverage or latency, within the declared budget.

Table III reports the frozen 20-seed holdout, Table IV reports results stratified by onset regime, and Table V reports the complete admissibility outcomes. Higher recall is preferred, whereas lower TTA_{ef}, FAR, SAR, and redundant-alert counts are preferred. Alerts per trip must remain within the budget $B{=}2.5$.

TABLE III
FROZEN 20-SEED HOLDOUT (240 PERSONA×SEED CLUSTERS). TTA_{ef} IN SECONDS; BUDGET $B{=}2.5$ ALERTS/TRIP. A0 IS DESCRIPTIVE; REDUNDANT RE-ALERTS ARE DEFINED WITHIN GOVERNED EVENTS AND DO NOT APPLY TO A0.

Arm	Recall	TTA_{ef}	FAR/tr.	Alerts/tr.	SAR	Red./tr.
A0 Legacy	1.000	1.49	1.250	3.486	0.034	—
A1 EGS	0.825	4.14	0.874	2.898	0.030	1.33
A2 EAML	**0.846**	4.21	**0.335**	**2.481**	**0.000**	**1.16**
A3 EAML−OSSL	0.830	4.42	0.321	2.539	0.007	1.22

TABLE IV
HOLDOUT STRATIFIED BY ONSET REGIME (PER ARM). PAIRED DIFFERENCES IN THE TEXT USE UNROUNDED CLUSTER-LEVEL VALUES; DISPLAYED MEANS ARE ROUNDED.

	Recall		TTA_{ef} (s)	
Arm	abrupt	progres.	abrupt	progres.
A0 Legacy	1.000	1.000	0.32	2.66
A1 EGS	0.834	0.815	3.12	5.16
A2 EAML	**0.852**	**0.840**	3.15	5.26
A3 EAML−OSSL	0.838	0.823	3.37	5.48

The comparison should not be interpreted as an optimization of recall alone. A0 illustrates this distinction. The legacy trigger reaches recall 1.0 and has the lowest raw latency, but produces 3.49 alerts per trip, which is 39% above the declared budget, with FAR 1.25. It therefore provides a descriptive reference rather than an admissible operating point under the study constraints.

Among the governed designs, A2 satisfies the complete predeclared admissibility specification relative to A1. Event recall improves by $+0.021$ (95% CI $[0.006, 0.037]$), while effective latency remains within the non-inferiority margin at $+0.07\,\text{s}$ (CI $[-0.16, 0.29]$). Alert load decreases from 2.90 to 2.48 per trip, placing A2 within the declared budget.

The alert-quality measures also improve. FAR decreases from 0.87 to 0.33 per trip, a 62% reduction (CI $[-0.57, -0.51]$). Redundant re-alerts decrease by 0.18 per trip, and SAR decreases from 0.030 to zero (CI of the paired difference $[-0.033, -0.027]$). No alert delivered by A2 in the 20-seed holdout satisfies the stale-alert criterion of Eq. (2). Section V-D later examines whether this zero-SAR result depends on the accounting threshold.

The stratified results show that the aggregate effect is not driven by one onset regime. Recall improves by $+0.018$ (CI $[0.002, 0.034]$) for abrupt episodes and by $+0.024$ (CI $[0.005, 0.044]$) for progressive episodes. The additional effective latency introduced by the mediation remains below 0.11 s in both regimes. Relative to A0, A2 also reduces FAR by 73% while satisfying the budget that A0 violates.

B. RQ2: role of suppression evidence

RQ2 tests whether the complete mediation remains effective when the suppression-evidence mechanism is removed. The ablation is evaluated in two ways: whether the resulting design remains admissible and whether stale-alert behavior returns. A3 removes both the Suppression Summary and the forced-refresh channel and therefore does not, by itself, isolate

TABLE V

PRE-DECLARED GATE OUTCOMES ON THE FROZEN HOLDOUT (PAIRED VS. A1; 240 CLUSTERS). U90 DENOTES THE UPPER ENDPOINT OF THE TWO-SIDED 90% INTERVAL (MEAN $+1.645$ SE) AND BRACKETS DENOTE 95% CIS. G2 IS EVALUATED IN AGGREGATE AND SEPARATELY FOR ABRUPT AND PROGRESSIVE REGIMES. THE NINE REPORTED CHECKS COMPRISE G1, THREE G2 CHECKS, G3, G4, AND THE THREE G5 QUALITY DIMENSIONS. G5 REQUIRES IMPROVEMENT IN AT LEAST ONE DIMENSION; 3/3 DENOTES IMPROVEMENT IN ALL THREE. DIFFERENCES IN THE TEXT USE UNROUNDED CLUSTER-LEVEL VALUES.

		A2 (EAML) vs. A1			**A3 (EAML−OSSL) vs. A1**		
Gate	Criterion	value	U90	verdict	value	U90	verdict
G1	$\mathrm{U90}(\Delta\mathrm{TTA}_{\mathrm{ef}}) < +1.0\,\mathrm{s}$	$+0.066$	0.254	pass	$+0.284$	0.471	pass
G2	ΔRecall ≥ -0.02 (aggregate)	$+0.021\,[0.006, 0.037]$	—	pass	$+0.006\,[-0.009, 0.021]$	—	pass
	abrupt / progressive	$+0.018$ / $+0.024$	—	pass	$+0.004$ / $+0.008$	—	pass
G3	Alerts/trip ≤ 2.5	2.481	—	pass	2.539	—	**fail**
G4	$\mathrm{U90}(\Delta\mathrm{FAR}) \leq 0$	-0.539	-0.514	pass	-0.553	-0.529	pass
G5	superiority ≥ 1 quality dim. (CI excl. 0)	SAR $-0.030\,[-0.033, -0.027]$	—	pass (3/3)	SAR $-0.023\,[-0.027, -0.020]$	—	pass
	Admissible			**yes (all gates)**			**no (G3)**

their individual contributions. Section V-D reports A4, which retains suppression evidence while disabling forced refresh, to decompose these two roles.

A3 fails the first test. Without OSS_t, reused probabilities enter the window without their associated quality and decay information, and the force channel is disabled. The resulting alert rate is 2.539 per trip, which exceeds the budget of 2.5 and fails G3. A3 is therefore inadmissible under the pre-declared specification.

The second effect appears in evidence freshness. Relative to A2, SAR increases from zero to 0.0066 (CI $[0.005, 0.008]$), and effective latency increases by $+0.22\,\mathrm{s}$ (CI $[0.17, 0.27]$). Under the evaluated design, removing the suppression-evidence contract therefore reintroduces the stale-alert behavior targeted by the OSSL and also worsens effective latency.

These results provide evidence that the evidence-aware design contributes to the observed admissibility and staleness outcomes under the evaluated simulation assumptions.

C. RQ3: oscillating-risk stress analysis (A-OSC)

RQ3 examines how the frozen designs behave when suppression and alert timing interact under an oscillating-risk regime that is absent from the confirmatory holdout. This stress analysis was declared in the second protocol amendment before holdout inspection. It re-executes the four frozen arms, without re-selection, on ten new seeds containing two additional oscillatory recipes: borderline risk fluctuating ± 0.12 around the guard-opening threshold and critical episodes with pre-onset fluctuation.

The stress analysis has no admissibility gates. Its purpose is to characterize how the designs behave outside the nominal evaluation regime.

The four arms respond differently to the oscillatory dynamics. A0 produces 13.1 alerts per trip, corresponding to $5.2\times$ the declared budget, with FAR 9.9. Such a high false and repeated alert load is consistent with conditions associated with alert fatigue [14], [15].

A1 exhibits the opposite trade-off. Oscillation around θ_{on} repeatedly opens and closes governed events, consuming the per-trip cap early. Its recall decreases to 0.296, with $\mathrm{TTA}_{\mathrm{ef}} = 11.7\,\mathrm{s}$.

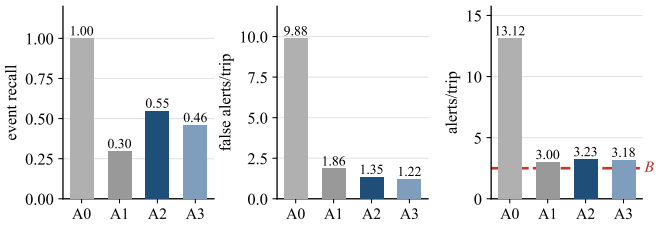


Fig. 2. A-OSC stress regime (10 fresh seeds, frozen arms, no re-selection). A0 produces 13.1 alerts per trip, A1 falls to 0.296 recall, and A2 retains substantially more coverage than A1 while producing fewer alerts than A0. The dashed line marks the alert budget B=2.5, which no arm satisfies in this regime.

A2 retains substantially more coverage than A1. Relative to A1, recall increases by $+0.252$ (CI $[0.228, 0.277]$), effective latency improves by 3.5 s (CI $[-3.88, -3.16]$), and FAR decreases by 0.51 per trip. SAR remains zero. Maintaining fresh evidence under this regime requires more detector executions: the force channel is activated 5.9 times per trip, compared with 0.39 in the confirmatory holdout.

A3 again shows the effect of removing the complete evidence-aware mechanism. Its SAR increases to 0.108 (A3 vs. A2 difference CI $[0.097, 0.120]$), and recall is 0.089 lower than A2.

None of the four arms satisfies the 2.5-alert budget in A-OSC. A2 produces 3.23 alerts per trip and retains only part of A0's coverage. The stress analysis therefore does not establish robustness to arbitrary oscillation. It shows that A2 preserves better evidence freshness and substantially more coverage than A1 under this particular stress regime, while still exhibiting a measurable budget violation.

D. Post-hoc mechanism decomposition and baseline checks

Three post-hoc analyses examine alternative explanations for the confirmatory results. These analyses use no admissibility gates and do not alter the frozen A2 configuration. Table VI summarizes their main findings.

Mechanism decomposition with A4. A4 separates the two mechanisms that A3 removes together, retaining the complete

TABLE VI
POST-HOC MECHANISM AND BASELINE-ROBUSTNESS CHECKS (FROZEN A2 CONFIGURATION, NO RE-SELECTION).

Check	Main finding
A4 (OSS_t kept, force off)	behavior $\approx$ A2, but SAR returns to 0.0033 $[0.0023, 0.0044]$; 0.71 unfulfilled freshness requests/trip
SAR accounting threshold	A2 SAR $= 0$ at $0.50/0.60/0.70$ (11,907 alerts); A1 delivers 11.7% above 0.50
Re-optimized EGS (fresh seeds)	best pure safeguard: 3.17 alerts/trip ($> B$), FAR 0.66 vs. 0.36, SAR 0.018 vs. 0

OSSL evidence contract while disabling forced refresh. The EAML can therefore observe stale evidence and withhold an alert, but it cannot request a new detector execution.

On the same 20 holdout seeds, A4 does not differ detectably from A2 in recall, latency, false alerts, or alert budget, with all paired confidence intervals containing zero. Evidence freshness differs, however. SAR increases from zero in A2 to 0.0033 in A4 (CI $[0.0023, 0.0044]$), and A4 produces 0.71 unresolved freshness requests per trip. The same qualitative pattern is observed in A-OSC.

These results help separate the two roles. Retaining the OSSL evidence contract is consistent with the observed admissibility, budget, latency, and false-alert outcomes, while the force channel is associated with eliminating the remaining stale-alert cases through fresh computation before delivery.

Sensitivity to the SAR accounting threshold. To assess whether the zero-SAR result depends on the chosen threshold of 0.60, we recomputed SAR for accounting thresholds of 0.50, 0.60, and 0.70. Across the 11,907 alerts delivered by A2, none has a reused ratio above 0.50. SAR therefore remains zero under all three thresholds. In contrast, 11.7% of A1 alerts exceed 0.50 and 3.0% exceed 0.60. Zero SAR is therefore interpreted as empirical satisfaction of the freshness-restoration invariant under the evaluated scenarios, rather than as independent evidence of real-world safety or deployment readiness.

Robustness to baseline calibration. We also tested whether the advantage of A2 could be explained by the pre-existing calibration of A1. The event-governed safeguard was re-optimized over the same guard grid using the same development seeds, and the resulting frozen design was evaluated with A2 on ten new seeds (6001–6010).

No pure-safeguard configuration in the grid satisfies the 2.5-alert budget on the development data. The selected configuration produces 3.17 alerts per trip on the new seeds, a 27% budget violation, with FAR 0.66 compared with 0.36 for A2 and SAR 0.018 compared with zero. Under the declared alert budget, the re-optimized safeguard therefore remains inadmissible. Among the governed families evaluated in this study, A2 remains the only admissible design.

E. Runtime cost of the evidence layers

After evaluating alert quality, we measure the computational overhead introduced by the evidence-preservation and mediation layers. Table VII reports a microbenchmark of the three edge levels with detector inference excluded. The benchmark

TABLE VII
PER-FRAME OVERHEAD OF THE EVIDENCE LAYERS (PURE PYTHON, SINGLE X86-64 THREAD, n=200,000 FRAMES). THE DETECTOR IS A STUB; COSTS ARE THE ARCHITECTURE'S OWN OVERHEAD.

Component	p50 (μs)	p95 (μs)	Size
MHEG gate decision	0.8	1.4	—
OSSL OSS_t materialization	2.4	2.6	402 B/OSS_t
EAML window step (RISK$\wedge$GUARD)	39.4	59.0	$\leq$4.9 KB (window)
Full pipeline (stub detector)	44.8	68.6	—

processes 200,000 frames after 20,000 warm-up frames using a single-threaded pure-Python reference implementation on a desktop-class x86-64 core.

Producing one OSS_t requires 2.4 μs at the median. A complete EAML step, including the window update and RISK$\wedge$GUARD decision, requires 39 μs, while the complete per-frame architecture with a stub detector remains below 69 μs at the 95th percentile. A serialized OSS_t occupies 402 bytes, and the full EAML window occupies at most 4.9 KB.

At the evaluated frame rate of 9.6 fps, corresponding to approximately 104 ms per frame, the architecture's own processing overhead is below 0.07% of the frame-time budget. A tenfold slowdown would still keep this overhead below 1 ms per frame. These measurements should not be interpreted as embedded-hardware results. They come from unoptimized interpreted code on x86-64; energy consumption and latency on Raspberry-Pi- or Jetson-class platforms remain to be measured.

VI. DISCUSSION

Suppression evidence as an architectural requirement. Selective computation reduces resource use by avoiding detector executions on frames considered redundant. The results of this study show that this decision also affects the information available to downstream alerting. When a frame is skipped, an alerting component may need to know which state is being reused, how long it has been reused, how the input has changed since the last computation, and whether fresh inference should be requested before an alert is delivered. We refer to this residual obligation as the *evidence debt* of selective computation.

The OSSL makes this information explicit through its per-frame evidence contract. In the confirmatory holdout, the complete architecture produced no alerts whose evidence windows satisfied the stale-alert criterion. When the suppression evidence was removed in A3, stale alerts returned and the resulting design also exceeded the alert budget. The finer A4 analysis further showed that retaining suppression evidence while disabling forced refresh preserved most operational outcomes but no longer produced an exact-zero SAR. Together, these results support the role of both evidence preservation and freshness restoration in the evaluated architecture.

This finding should be interpreted as an architectural result rather than as evidence that the specific OSSL representation is universally optimal; whether the same evidence fields, thresholds, and mediation rules are appropriate in other systems requires separate evaluation.

Why the decision levels are separated. The experimental arms illustrate the consequences of assigning different responsibilities to the same mechanism. A0 directly links frame-level detections to alerts. Under oscillating risk, repeated threshold crossings therefore result in a high alert rate. A1 introduces event-governed delivery, but its alert decisions still lack explicit information about suppressed computation. Under the A-OSC regime, repeated event transitions consume the available alert capacity early and substantially reduce coverage.

A2 separates these responsibilities. The MHEG schedules computation, the OSSL represents suppression, and the EAML makes the delivery decision from accumulated evidence. The EAML can also request fresh computation through the MHEG before an alert is delivered. The results suggest that this separation is useful because computation scheduling, evidence quality, and alert delivery respond to different operational constraints.

Trade-off introduced by freshness restoration. The zero-SAR result does not come without additional computation. Freshness restoration required 0.39 forced computations per trip in the confirmatory holdout and 5.9 per trip in the oscillating-risk analysis. The increased use of the force channel under stress is consistent with its intended role: more uncertain or reuse-dominated evidence results in more requests for fresh inference.

The comparison with A0 also shows why raw latency alone cannot characterize alert quality: its lower latency comes with 3.49 alerts per trip (39% over budget) and substantially higher FAR. Among the governed designs, A2 and A1 operate at similar effective latency, while A2 reduces false alerts, redundant alerts, and stale alerts and remains within the alert budget. The relevant trade-off in this study is therefore between responsiveness, alert burden, and evidence quality rather than latency alone.

Role of explicit admissibility criteria. The amendment trail shows that the acceptance criteria were capable of rejecting candidate designs: two specifications were rejected with empty admissible sets before the final protocol adopted non-inferiority requirements together with explicit quality and budget constraints. These earlier empty sets are retained because they clarify how the final hypothesis was formed and demonstrate that acceptance was not guaranteed by the evaluation procedure.

Scope and transferability. The architecture was evaluated for vehicular risk communication, but the underlying problem is not specific to driving. Other edge systems may also suppress expensive computation while later making decisions from reused inference. In such systems, preserving information about skipped computation may help downstream components distinguish fresh evidence from reused evidence.

Different applications would require their own detectors, evidence-quality definitions, budgets, thresholds, and validation procedures. The transferable claim is therefore limited to the architectural principle that suppression and downstream decision making should not be treated as independent when the latter depends on the freshness of the former.

VII. Threats to Validity

Construct validity. The evaluation measures properties of risk communication rather than post-alert driving outcomes. Event recall, effective latency, FAR, redundant alerts, and SAR characterize whether and how alerts are delivered, but they do not measure whether the driver subsequently responds correctly or whether an unsafe event is prevented. The synthetic driver-response model is used only as part of the experimental dynamics and should not be interpreted as a validated behavioral model.

The alert budget, the non-inferiority margins, and the 0.60 reuse threshold are operational definitions used by this study rather than universal human-factors constants. Their provenance is reported explicitly in Sections III and IV.

Internal validity. All arms use the same episode generator, observation model, MHEG configuration, and predrawn exogenous realizations. Common random numbers are verified programmatically, and inference is based on paired persona$\times$seed differences. Development and holdout seeds are disjoint, and the confirmatory holdout was executed once after the configuration and final admissibility specification were frozen.

The freezing procedure is programmatic rather than externally preregistered. In addition, the $1.0\,\mathrm{s}$ latency non-inferiority margin was chosen after development-data inspection. The methodological protection is instead the separation between development and the untouched holdout, together with the preserved amendment trail.

A1 is a previously frozen event-governed baseline and was not originally optimized for the current experiment. This could create a calibration advantage for A2. The post-hoc baseline check addresses this concern by re-optimizing the safeguard using the same development seeds and testing the resulting configuration on fresh seeds. No evaluated pure-safeguard configuration satisfies the declared alert budget, although this result is limited to the explored parameter grid and generator.

External validity. All behavioral episodes and driver personas are synthetic. Feature and risk calibration draw on naturalistic-driving statistics, but the observation and response processes are declared models rather than models learned from naturalistic trajectories. The reported results therefore characterize the evaluated synthetic environment and should not be interpreted as estimates of real-world driver behavior.

The personas encode risk, responsiveness, and fatigue parameters for experimental variation only and support no demographic or population claims. Similarly, A-OSC is a deliberately constructed stress regime designed to exercise interactions between suppression and alert governance. It represents one out-of-nominal condition rather than a realistic distribution-shift model.

Implementation validity. The computational-cost measurements were obtained from a single-threaded pure-Python reference implementation on a desktop-class x86-64 processor. They isolate the overhead of the proposed architecture and exclude the cost of the detector itself. Although the measured

overhead is small relative to the evaluated frame period, these results are not equivalent to measurements on automotive or embedded hardware. Runtime latency, energy consumption, memory behavior, and thermal constraints on Raspberry-Pi-, Jetson-, or automotive-grade platforms remain to be measured.

VIII. Conclusion

We presented a cognitive edge architecture that separates selective computation, suppression-evidence preservation, episodic alert delivery, and slower configuration adaptation. Its central mechanism is a per-frame evidence contract that makes skipped computation observable to the alerting layer, together with a single mediated origin for driver alerts.

Under the frozen 20-seed confirmatory holdout, the complete architecture satisfied all pre-declared admissibility gates relative to the event-governed baseline. Event recall increased by 0.021 at effective latency parity, false alerts per trip decreased by 62%, and SAR decreased from 0.030 to zero. Removing the suppression-evidence mechanism made the resulting design inadmissible and reintroduced stale alerts.

The oscillating-risk analysis showed a different operating regime. A0 produced 13.1 alerts per trip, A1 recall decreased to 0.296, and A2 retained substantially more coverage than A1 while empirically satisfying the zero-stale-alert invariant under the evaluated accounting thresholds. However, A2 also exceeded the alert budget in this regime. These results therefore indicate improved evidence quality and alert behavior under the evaluated stress condition rather than immunity to out-of-nominal dynamics.

Within the synthetic environment considered here, the results support a specific architectural principle: when an edge system suppresses computation, downstream communication decisions should have explicit access to information about the resulting evidence freshness. Future work will evaluate this principle using naturalistic driving data, measure its computational and energy cost on embedded hardware, and test the evidence contract in a second application domain.

Artifact Availability

The artifact contains the episode generator, behavioral recipes, synthetic personas, implementations of the evaluated arms, all three protocol specifications, the two empty-set development outcomes, the frozen confirmatory configuration and holdout outputs, the A-OSC stress analysis, mechanism-decomposition and baseline-robustness results, the runtime microbenchmark, and a SHA-256 integrity manifest. An anonymized artifact containing the code, frozen configurations, seeds, raw outputs, table-generation scripts, and SHA-256 manifest is provided as supplementary material. The archival repository and DOI will be disclosed in the camera-ready version.

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Anthropic Claude (Fable 5) assisted with language editing throughout Sections I–VIII and with the development and review of the experimental code. The research questions, architecture, experimental protocol, implementation decisions, result validation, scientific interpretation, and conclusions remained with the authors. All AI-assisted outputs were reviewed by the authors, who retain final responsibility for the manuscript.

References

[1] W. Shi, J. Cao, Q. Zhang, Y. Li, and L. Xu, "Edge computing: Vision and challenges," *IEEE Internet Things J.*, vol. 3, no. 5, pp. 637–646, 2016.

[2] S. Liu *et al.*, "Edge computing for autonomous driving: Opportunities and challenges," *Proc. IEEE*, vol. 107, no. 8, pp. 1697–1716, 2019.

[3] V. Campos, B. Jou, X. Giró-i-Nieto, J. Torres, and S.-F. Chang, "Skip RNN: Learning to skip state updates in recurrent neural networks," in *Proc. ICLR*, 2018.

[4] A. Graves, "Adaptive computation time for recurrent neural networks," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2016.

[5] Z. Wu, C. Xiong, C.-Y. Ma, R. Socher, and L. S. Davis, "AdaFrame: Adaptive frame selection for fast video recognition," in *Proc. IEEE/CVF CVPR*, 2019, pp. 1278–1287.

[6] Z. Wu, C. Xiong, Y.-G. Jiang, and L. S. Davis, "LiteEval: A coarse-to-fine framework for resource efficient video recognition," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2019.

[7] M. Xu, Y. Xiong, H. Chen, X. Li, W. Xia, Z. Tu, and S. Soatto, "Long short-term transformer for online action detection," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2021, pp. 22 610–22 622.

[8] F. Sajid, A. R. Javed, A. Basharat, N. Kryvinska, A. Afzal, and M. Rizwan, "An efficient deep learning framework for distracted driver detection," *IEEE Access*, vol. 9, 2021, doi: 10.1109/ACCESS.2021.3138137.

[9] M. Fresta, F. Bellotti, I. Bochenko, L. Lazzaroni, G. Merlhiot, F. Tango, and R. Berta, "Deep learning-based real-time driver cognitive distraction detection," *IEEE Access*, vol. 13, pp. 26 589–26 607, 2025, doi: 10.1109/ACCESS.2025.3539392.

[10] J. O. Kephart and D. M. Chess, "The vision of autonomic computing," *Computer*, vol. 36, no. 1, pp. 41–50, 2003.

[11] D. Weyns, *An Introduction to Self-Adaptive Systems: A Contemporary Software Engineering Perspective*. Wiley–IEEE Press, 2020.

[12] D. Seto, B. Krogh, L. Sha, and A. Chutinan, "The Simplex architecture for safe online control system upgrades," in *Proc. ACC*, vol. 6, 1998, pp. 3504–3508.

[13] M. Alshiekh, R. Bloem, R. Ehlers, B. Könighofer, S. Niekum, and U. Topcu, "Safe reinforcement learning via shielding," in *Proc. AAAI*, 2018, pp. 2669–2678.

[14] G. Abe and J. Richardson, "Alarm timing, trust and driver expectation for forward collision warning systems," *Appl. Ergon.*, vol. 37, no. 5, pp. 577–586, 2006.

[15] J. P. Bliss and S. A. Acton, "Alarm mistrust in automobiles: How collision alarm reliability affects driving," *Appl. Ergon.*, vol. 34, no. 6, pp. 499–509, 2003.

[16] T. A. Dingus, F. Guo, S. Lee, J. F. Antin, M. Perez, M. Buchanan-King, and J. Hankey, "Driver crash risk factors and prevalence evaluation using naturalistic driving data," *Proc. Natl. Acad. Sci. U.S.A.*, vol. 113, no. 10, pp. 2636–2641, 2016.

[17] A. M. Law, *Simulation Modeling and Analysis*, 5th ed. McGraw-Hill, 2015.

[18] R. D. Yates, Y. Sun, D. R. Brown III, S. K. Kaul, E. Modiano, and S. Ulukus, "Age of information: An introduction and survey," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 5, pp. 1183–1210, 2021.

[19] M. K. C. Shisher and Y. Sun, "How does data freshness affect real-time supervised learning?" in *Proc. ACM MobiHoc*, 2022.

[20] I. Gokarn, H. Sabbella, Y. Hu, T. Abdelzaher, and A. Misra, "MOSAIC: Spatially-multiplexed edge AI optimization over multiple concurrent video sensing streams," in *Proc. ACM MMSys*, 2023.

[21] J. Zhu and J. Gong, "Multi-source peak age of information optimization in mobile edge computing systems," *IEEE/ACM Trans. Netw.*, 2025.

[22] V.-P. Bui, S. R. Pandey, P. M. de Sant Ana, B. Soret, and P. Popovski, "Timely communication from sensors for wireless networked control in cloud-based digital twins," arXiv preprint arXiv:2408.10241, 2024.

[23] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, pp. 1735–1780, 1997.